

INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

Department of Computer and Communication Engineering

Course Code: CCE-3509

Course Title: Computer Architecture

Abu Noman Siddike
Adjunct Faculty
Dept. of CCE, IIUC

COURSE OVERVIEW

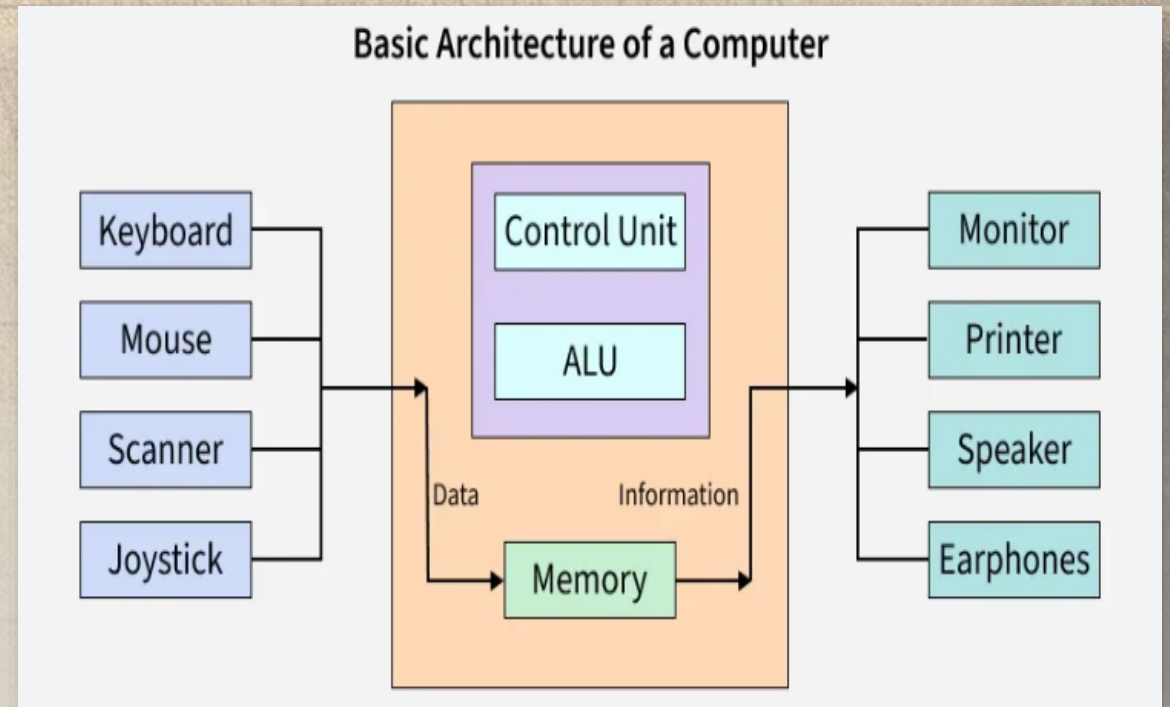
- Credit Hour: 3
- Contact Hour: 3 period per week
- Mid: 30 marks
- Final: 50 marks
- Assignment: 10 marks
- Attendance: 10 marks

COMPUTER ARCHITECTURE

- **What is Computer Architecture?**

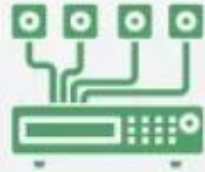
Computer Architecture is the science and art of selecting and interconnecting hardware components to create computers that meet functional, performance and cost goals.

- **Computer Architecture** is the *blueprint* that defines how a computer's parts are organized and how they work together to execute instructions.
- It bridges the gap between **abstract computing concepts** (like algorithms and programming languages) and the **physical hardware** that runs them.



COMPONENTS OF COMPUTER ARCHITECTURE

Components of Computer Architecture



Input unit and
associated peripherals



Output unit and
associated peripherals



Storage unit/
memory



Central processing
unit (CPU)



Operating system
(OS)

ARCHITECTURE VS ORGANIZATION

	Computer Architecture	Computer Organization
Purpose	Computer architecture explains what a computer should do.	Computer organization explains how a computer works.
Target	Computer architecture provides functional behavior of computer system.	Computer organization provides structural relationships between parts of computer system.
Design	Computer architecture deals with high level design.	Computer organization deals with low level design.
Actors	Actors in Computer architecture are hardware parts.	Actor in computer organization is performance.
Order	Computer architecture is designed first.	Computer organization is started after finalizing computer architecture.

VON NEUMANN VS HARVARD ARCHITECTURE

Feature	Von Neumann	Harvard
Memory Structure	Single memory stores both instructions and data	Separate memories for instructions and data
Buses	One shared bus for data and instructions → can cause the Von Neumann bottleneck	Separate buses for data and instructions → allows simultaneous access
Instruction/Data Access	Cannot fetch instruction and read/write data at the same time	Can fetch instruction and access data in parallel
Complexity	Simpler design, easier and cheaper to implement	More complex due to dual memory systems
Performance	Slower in some cases due to bus contention	Faster for certain workloads because of parallelism
Flexibility	More flexible - programs and data share the same space, easy to modify	Less flexible — fixed separation of instruction/data memory
Typical Use	General-purpose computers, desktops, laptops	Embedded systems, DSPs, microcontrollers

COMPUTER EVOLUTION AND PERFORMANCE

Computer Generations	Periods	Based on	Properties	Example
1 st	1940-1956	Vacuum tubes	Huge, power-hungry, slow, expensive	ENIAC
2 nd	1956-1963	Transistor	Smaller, faster, more reliable, less heat;	IBM 1401, PDP-1
3 rd	1964-1971	Integrated Circuit (ICs)	Dramatic size & cost reduction, better speed, standardization	IBM 360,370 series,
4 th	1971-present	Microprocessor	Personal computing era, exponential performance growth (Moore's Law)	Intel 4004 → modern CPUs
5 th	Present and Beyond	AI (Artificial Intelligence)	Specialized architectures, massive parallel processing, domain-specific speedups	Modern desktop, laptop, macbooks etc.

COMPUTER EVOLUTION

- Hardware Miniaturization:

Vacuum Tube > Transistor > IC > Microprocessor

Smaller components = faster clock speed

- Architectural Innovation: Pipelining, Caching, Superscalar execution
- Parallelism: Multicore CPUs, GPUs
- Specialized Processing: AI accelerators (TPUs, NPUs)

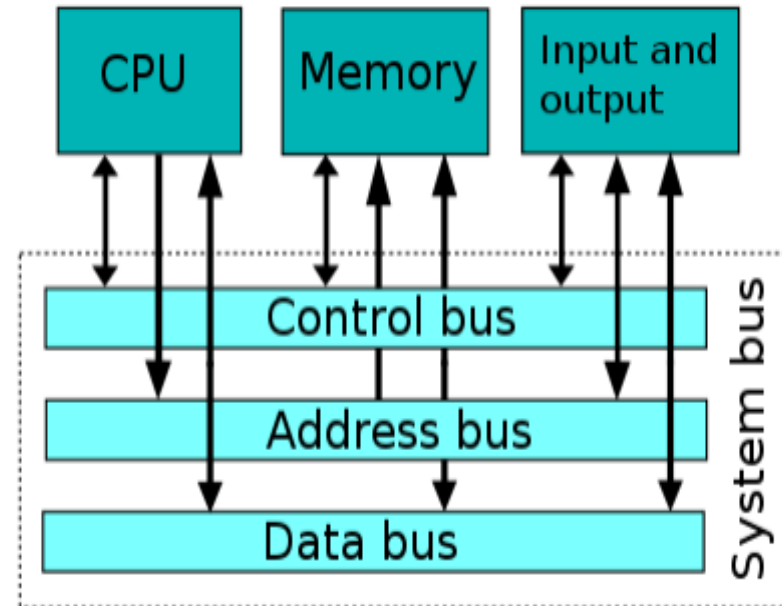
BUS

A computer bus is a communication system used to transfer data between components within a computer or between different computers.

- It consists of physical connections like wires, circuits, or cables.
- Components like the CPU, memory, and input/output (I/O) devices are connected through a bus.
- It simplifies data transfer and improves efficiency.

TYPES OF BUSES

- ✓ **Address bus** - carries memory addresses from the processor to other components such as primary storage and input/output devices. The address bus is unidirectional.
- ✓ **Data bus** - carries the data between the processor and other components. The data bus is bidirectional.
- ✓ **Control bus** - carries control signals from the processor to other components. The control bus also carries the clock's pulses. The control bus is unidirectional.



BUS

Bus Type	Direction	Purpose	Key Role
Address Bus	Unidirectional	Carries memory addresses	Identifies where data should go
Data Bus	Bidirectional	Carries actual data	Moves data between components
Control Bus	Bidirectional	Carries control and sync signals	Coordinates CPU and device actions

The background of the image features faint, sepia-toned sketches of two airships. On the left is a hot air balloon with a large, ornate, spherical envelope decorated with various patterns and a small basket hanging below. On the right is a rigid blimp or airship with a long, cylindrical hull, a series of vertical struts supporting a lower deck, and a small tail fin. The text "The End" is written in a red, cursive script across the center of the image.

The End

